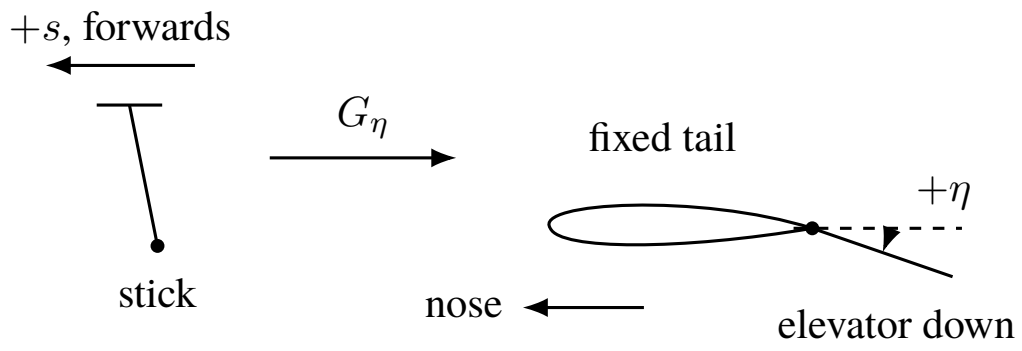
PART 2 / STICK GEARING AND FORCE

From elevator angle to stick movement and force

Part 1 supplied the elevator and hinge-coefficient trim gradients. Start by connecting the elevator to the pilot's stick. Take s positive forwards and η positive for elevator trailing edge down. With suitable reference origins:

$$\eta = G_\eta s, \qquad d\eta = G_\eta \, ds, \qquad \frac{ds}{d\eta} = \frac{1}{G_\eta}.$$

G_η is the stick gearing in $\mathrm{rad/m}$; $G_\eta > 0$.



Let P_η be the pilot force, positive for a push. The control linkage supplies holding torque $\tau_{\mathrm{hold}} = -H$, opposing the aerodynamic hinge torque H. Equate the work transmitted through an ideal reversible linkage:

$$P_\eta \, ds = \tau_{\mathrm{hold}} \, d\eta = \tau_{\mathrm{hold}} G_\eta \, ds.$$

$$P_\eta = G_\eta \tau_{\mathrm{hold}} = -G_\eta H.$$

The work is in joules, torque in newton metres and force in newtons. In magnitude, $|P_\eta| = G_\eta |H|$. This lecture uses the ideal linkage model, without friction or artificial feel.

1 / LIFT COEFFICIENT AND SPEED

Differentiate the steady level-flight balance

Compare trimmed level-flight conditions at the same weight, density and configuration:

$$L = W = \frac{1}{2}\rho U_\infty^2 S_W C_L.$$

U_∞ is the air-relative flight speed, written V in Week 4. Rearrange before differentiating:

$$U_\infty^2 C_L = \frac{2W}{\rho S_W} = \text{constant}.$$

Apply the product rule:

$$\begin{aligned} d(U_\infty^2 C_L) &= 0, \\ 2U_\infty C_L\, dU_\infty + U_\infty^2\, dC_L &= 0, \\ U_\infty^2\, dC_L &= -2U_\infty C_L\, dU_\infty. \end{aligned}$$

Divide by $U_\infty^2\, dU_\infty$:

$$\frac{dC_L}{dU_\infty} = -\frac{2C_L}{U_\infty}.$$

Higher speed gives more dynamic pressure. The same weight can therefore be supported with a smaller lift coefficient. This slope has units $\mathrm{s/m}$. Weight and density remain constant during this differentiation.

2 / STICK MOVEMENT AND SPEED

Apply the chain rule to the stick position

Speed changes the required C_L. That changes the trimmed elevator angle, which changes the stick position:

$$U_\infty \longrightarrow C_L \longrightarrow \eta_{\text{trim}} \longrightarrow s.$$

$$\frac{ds}{dU_\infty} = \frac{ds}{d\eta}\frac{d\eta_{\text{trim}}}{dC_L}\frac{dC_L}{dU_\infty}.$$

Insert each factor separately:

$$\frac{ds}{d\eta} = \frac{1}{G_\eta}, \qquad \frac{d\eta_{\text{trim}}}{dC_L} = -\frac{H_n}{\bar{V}_T a_2}, \qquad \frac{dC_L}{dU_\infty} = -\frac{2C_L}{U_\infty}.$$

$$\begin{aligned}\frac{ds}{dU_\infty} &= \frac{1}{G_\eta}\left(-\frac{H_n}{\bar{V}_T a_2}\right)\left(-\frac{2C_L}{U_\infty}\right) \\ &= \frac{2C_L}{G_\eta U_\infty \bar{V}_T a_2} H_n.\end{aligned}$$

At the reference equilibrium, denoted by subscript e:

$$\left.\frac{ds}{dU_\infty}\right|_e = \frac{2C_{L,e}}{G_\eta U_{\infty,e} \bar{V}_T a_2} H_n.$$

For an aft tail with positive fixed margin, the slope is positive: higher steady speed needs a more forward stick position. The derivative describes a sequence of equilibria, with the tab fixed.

3 / HINGE COEFFICIENT AND SPEED

Apply the chain rule to the hinge coefficient

Use the second trim gradient from Part 1:

$$\frac{dC_{H,\mathrm{trim}}}{dC_L} = -\frac{b_2 H'_n}{\bar{V}_T a_2}.$$

$$\begin{aligned}\frac{dC_{H,\mathrm{trim}}}{dU_\infty} &= \frac{dC_{H,\mathrm{trim}}}{dC_L}\frac{dC_L}{dU_\infty} \\ &= \left(-\frac{b_2 H'_n}{\bar{V}_T a_2}\right)\left(-\frac{2C_L}{U_\infty}\right) \\ &= \frac{2C_L b_2 H'_n}{U_\infty \bar{V}_T a_2}.\end{aligned}$$

$$H'_n > 0, \quad b_2 < 0, \quad \bar{V}_T, a_2 > 0 \quad \Longrightarrow \quad \frac{dC_{H,\mathrm{trim}}}{dU_\infty} < 0.$$

Convert the coefficient into aerodynamic hinge torque and pilot holding force:

$$H = qS_e\bar{c}_e C_H, \qquad P_\eta = -G_\eta q S_e \bar{c}_e C_H.$$

S_e and $\bar{c}_e$ are the elevator reference area and chord. At a force-free trim speed, $C_H = 0$; a small speed increase then requires a forward holding force for $H'_n > 0$. The moment and gearing method also applies to a canard when its signed arm and local-flow relation are used. The sign conclusions above assume a conventional aft tail.

4 / EXAMPLE 1

Calculate the two trim gradients

Question. Find $d\eta_{\text{trim}}/dC_L$ and $dC_{H,\text{trim}}/dC_L$ for the following aircraft:

$$H_n = 0.10, \quad H_n' = 0.05, \quad \bar{V}_T = 0.50,$$

$$a_2 = 1.0\,\text{rad}^{-1}, \qquad b_2 = -0.04\,\text{rad}^{-1}.$$

First use the fixed margin:

$$\begin{aligned}\frac{d\eta_{\text{trim}}}{dC_L} &= -\frac{H_n}{\bar{V}_T a_2} \\ &= -\frac{0.10}{0.50 \cdot 1.0} = \boxed{-0.20\,\text{rad}}.\end{aligned}$$

Then use the free margin:

$$\begin{aligned}\frac{dC_{H,\text{trim}}}{dC_L} &= -\frac{b_2 H_n'}{\bar{V}_T a_2} \\ &= -\frac{(-0.04)\cdot 0.05}{0.50 \cdot 1.0} = \boxed{4.0 \times 10^{-3}}.\end{aligned}$$

A larger required C_L needs more up-elevator and a larger aerodynamic hinge coefficient for these supplied signs. These are gradients per unit C_L. The next example converts them into gradients with respect to speed.

4 / EXAMPLE 2

Calculate stick movement and coefficient change

Question. For the same aircraft, use

$$\frac{dC_L}{dU_\infty} = -0.02\,\mathrm{s/m}, \qquad G_\eta = 1.0\,\mathrm{rad/m}.$$

Find ds/dU_∞ and $dC_{H,\mathrm{trim}}/dU_\infty$.

$$\begin{aligned}\frac{ds}{dU_\infty} &= \frac{1}{G_\eta}\frac{d\eta_{\mathrm{trim}}}{dC_L}\frac{dC_L}{dU_\infty} \\ &= \frac{1}{1.0}\cdot(-0.20)\cdot(-0.02) \\ &= \boxed{4.0\times10^{-3}\,\mathrm{s}}.\end{aligned}$$

$$\begin{aligned}\frac{dC_{H,\mathrm{trim}}}{dU_\infty} &= \frac{dC_{H,\mathrm{trim}}}{dC_L}\frac{dC_L}{dU_\infty} \\ &= (4.0\times10^{-3})\cdot(-0.02) \\ &= \boxed{-8.0\times10^{-5}\,\mathrm{s/m}}.\end{aligned}$$

A small $1\,\mathrm{m/s}$ increase in steady speed requires about 4 mm of forward stick movement.

The unit of ds/dU_∞ is $\mathrm{m/(m/s)} = \mathrm{s}$. It is displacement per speed change, rather than a response time.

Conclusions

Gearing connects the elevator angle to the stick displacement:

$$d\eta = G_\eta \, ds, \qquad P_\eta = G_\eta \tau_{\text{hold}} = -G_\eta H.$$

At fixed weight and density, steady level flight gives

$$\frac{dC_L}{dU_\infty} = -\frac{2C_L}{U_\infty}.$$

The two chains then give

$$\frac{ds}{dU_\infty} = \frac{2C_L H_n}{G_\eta U_\infty \bar{V}_T a_2},$$
$$\frac{dC_{H,\text{trim}}}{dU_\infty} = \frac{2C_L b_2 H'_n}{U_\infty \bar{V}_T a_2}.$$

For the stated aft-tail signs, a positive fixed margin gives forward stick movement as steady speed increases. A positive free margin gives a decreasing hinge coefficient and, near force-free trim, increasing forward holding force.

> Part 1 connects CG position to stability and trim gradients. Part 2 connects those gradients to the pilot's controls.

Week 6 extends the stability reasoning to sideslip, roll and yaw.